

Housing Burden & Insurance Expenditures

Portfolio Project — Melanie M. Lytell

Project Overview

This project examines the relationship between household housing burden and insurance expenditures. Using household-level Consumer Expenditure Survey data, the analysis investigates whether the share of household income devoted to mandatory housing expenses is associated with spending on four types of insurance: vehicle, life, personal, and health insurance.

The project uses regression analysis to examine these relationships while accounting for demographic, household, socioeconomic, geographic, and insurance-specific factors.

Research Question

How is housing burden associated with household spending on vehicle, life, personal, and health insurance?

The original hypothesis was that higher housing burden would be associated with lower insurance spending because households facing greater housing costs may have fewer resources available for other forms of financial protection. The analysis also examines whether the relationship differs across insurance categories and household characteristics.

Data

The analysis uses quarterly Consumer Expenditure Survey microdata from the U.S. Bureau of Labor Statistics, covering Q2 2018 through Q1 2024.

The regression sample contains 91,991 households across the United States.

The primary variable of interest is housing burden, defined as:

Mandatory housing expenditures divided by gross income.

Other variables include age, family size, housing tenure, work hours, marital status, race, sex, Hispanic origin, education, state, and quarter.

Data Preparation

The dataset was compiled and cleaned before the R analysis.

Housing burden observations below 0 or above 1 were removed. The insurance expenditure variables were adjusted to account for zero expenditures and then log transformed for use as the dependent variables in the regression models.

Categorical demographic variables were incorporated into the models as factors.

Methods

Four separate linear regression models were estimated:

- Vehicle insurance expenditures
- Life insurance expenditures
- Personal insurance expenditures
- Health insurance expenditures

Each model included housing burden as the primary explanatory variable along with demographic and household controls. State fixed effects and a quarterly time variable were also included. Additional controls were included where appropriate for the specific type of insurance.

The analysis evaluates the sign and statistical significance of the housing-burden coefficient and compares model fit across the four insurance categories.

Key Findings

Housing burden had different associations across the four types of insurance.

Insurance Type	Housing Burden Coefficient
Vehicle Insurance	2.127***
Life Insurance	0.482***

Personal Insurance	-0.742***
Health Insurance	-0.538***

The results show positive associations between housing burden and vehicle and life insurance expenditures, while personal and health insurance expenditures had negative associations with housing burden.

The findings therefore did **not** support the original hypothesis uniformly across all four insurance categories. Instead, the relationship varied by type of insurance.

Other Household Characteristics

The regression models also identified differences associated with household characteristics.

Age was positively associated with life and health insurance spending. Education was generally associated with greater insurance expenditures, particularly for personal insurance. Housing tenure, marital status, family size, race, sex, and Hispanic origin also showed relationships that varied depending on the type of insurance being examined.

These results demonstrate that insurance expenditures are associated with more than housing burden alone.

Visualizations

The project includes R-generated visualizations showing:

- The distribution of household housing burden
- The distribution of reference-person age
- Health insurance expenditures in relation to housing burden
- Housing burden coefficients across the four insurance models

The visualizations were designed in a simple black-and-white format for clear presentation.

Model Fit

The regression models produced different levels of explanatory power:

Model	R^2
Vehicle Insurance	0.127
Life Insurance	0.083
Personal Insurance	0.542
Health Insurance	0.248

These differences indicate that the included variables explain different proportions of variation in expenditures across the insurance categories.

Limitations

One important limitation is that the analysis identifies associations rather than causal effects. Housing burden may be related to other financial circumstances or characteristics that are not fully captured by the available variables.

Household preferences, financial behavior, and other factors may also influence insurance expenditures.

The study therefore provides evidence about relationships between housing burden and insurance expenditures rather than establishing that changes in housing burden directly cause changes in insurance spending.

Conclusion

This project examined how household housing burden is associated with spending on vehicle, life, personal, and health insurance using a large household-level dataset.

The results revealed different relationships across insurance categories. Housing burden was positively associated with vehicle and life insurance expenditures and negatively associated with personal and health insurance expenditures. The analysis also found that demographic and household characteristics were associated with insurance spending.

Overall, the project demonstrates how R, regression analysis, and data visualization can be used to investigate a real-world economic question using household-level data.

Tools

R — statistical analysis and visualization

Excel — data compilation and preparation

GitHub — project organization and portfolio presentation